

Task 1 - System Analysis Task

Think about the business. Extract the models. Assign the sources.

SOURCE TAG LEGEND

Tag	Name	Definition
System Generated	System Generated	The system assigns this value automatically. The user never types it.
User Input	User Input	The user types this value directly at the keyboard when prompted.
Default Value	Default Value	A fixed value set by the code when the object is created.
Calculated	Calculated	Derived automatically from another object already in memory.
From List	From List	The user selects a value from a list of options on screen.

Domain 1 — Library Management System

Entity 1: Member

Property Name	Data Type	Source Tag
MemberId	int	System Generated
Name	string	User Input
Email	string	User Input
PhoneNumber	string	User Input
RegistrationDate	DateTime	System Generated
IsActive	bool	Default Value

Entity 2: Book

Property Name	Data Type	Source Tag
BookId	int	System Generated
Title	string	User Input
Author	string	User Input
Genre	string	From List
TotalCopies	int	User Input
AvailableCopies	int	Calculated

Entity 3: Loan

Property Name	Data Type	Source Tag
LoanId	int	System Generated
MemberId	int	From List
BookId	int	From List
BorrowDate	DateTime	System Generated
DueDate	DateTime	Calculated
ReturnDate	DateTime	User Input
IsReturned	bool	Calculated

Entity 4: Reservation

Property Name	Data Type	Source Tag
ReservationId	int	System Generated
MemberId	int	From List
BookId	int	From List
ReservationDate	DateTime	System Generated
QueuePosition	int	Calculated
NotificationSent	bool	Default Value

Step 2 - Entity Relationships:

A member registers and gets assigned an ID. They can borrow books that are available. Each borrow creates a loan record. If a book is fully borrowed out, members can reserve it and the system notifies them. If a book gets returned late, a fine gets calculated. The loan gets closed when the book is returned.

Step 3 - Library Services:

#	Service Name	What it does
1	RegisterMember	Adds a new member to the system
2	BorrowBook	Records when a member takes out a book
3	ReturnBook	Marks a book as returned and calculates fines
4	ReserveBook	Lets a member reserve a book thats already checked out
5	ViewLoans	Shows all the books currently borrowed
6	ViewOverdue	Shows books that are overdue sorted by how late
7	PopularBooks	Shows which books are borrowed most

8	CalculateFine	Works out how much a member owes for late returns
9	CancelReserve	Removes a reservation from the queue
10	CheckActive	Checks if a member account is still active

Domain 2 — Hotel Management System

Guest

Property Name	Data Type	Source Tag
guestId	int	System Generated
guestName	string	User Input
guestPhone	string	User Input
guestEmail	string	User Input
guestNationality	string	User Input
registrationDate	string	System Generated

Room

Property Name	Data Type	Source Tag
roomId	int	System Generated
roomNumber	string	User Input
roomType	string	From List
floorNumber	int	User Input
nightlyPrice	decimal	Default Value
isAvailable	bool	Calculated

Booking

Property Name	Data Type	Source Tag
bookingId	int	System Generated
guestId	int	From List
roomId	int	From List
checkInDate	string	User Input
checkOutDate	string	User Input
numberOfNights	int	Calculated
totalPrice	decimal	Calculated
status	string	Default Value

CheckInRecord

Property Name	Data Type	Source Tag
recordId	int	System Generated
bookingId	int	From List
guestId	int	From List
roomId	int	From List
actualCheckIn	string	System Generated
actualCheckOut	string	System Generated
finalAmount	decimal	Calculated

Step 2 - Entity Relationships:

Guest registers and gets an ID. They make a booking for a room with check-in and check-out dates. The system figures out how many nights that is and calculates the total cost. When they check in, a record gets created with the actual time. When they check out, the room is marked available again. They can also cancel before checking in.

Step 3 - Hotel Services:

#	Service Name	What it does
1	RegisterGuest	Sign up a new guest
2	CreateBooking	Make a reservation for a room and dates
3	CheckIn	Mark when a guest arrives and create check-in record
4	CheckOut	Mark when a guest leaves and free up the room

5	CancelBooking	Cancel a reservation before check-in
6	AvailableRooms	Show what rooms are open
7	GuestHistory	Look up a guests past stays
8	GetPrice	Calculate what the stay costs
9	UpdateRoom	Change if a room is available or occupied